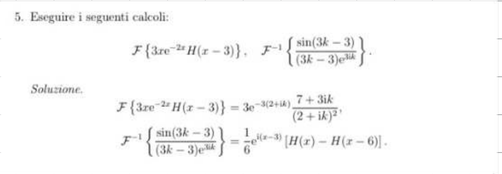
**ESERCIZIO n° 1**



5. Eseguire i seguenti calcoli:

$$\mathcal{F}\left\{3xe^{-2x}H(x-3)\right\}, \quad \mathcal{F}^{-1}\left\{\frac{\sin(3k-3)}{(3k-3)e^{3ik}}\right\}.$$

*Soluzione.*

$$\mathcal{F}\left\{3xe^{-2x}H(x-3)\right\} = 3e^{-3(2+ik)}\frac{7+3ik}{(2+ik)^2},$$

$$\mathcal{F}^{-1}\left\{\frac{\sin(3k-3)}{(3k-3)e^{3ik}}\right\} = \frac{1}{6}e^{i(x-3)}\left[H(x)-H(x-6)\right].$$

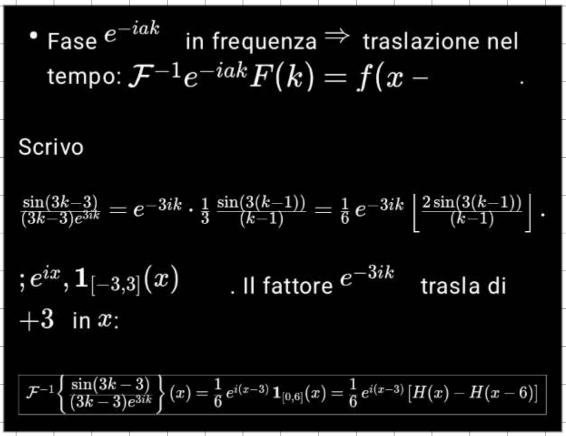
$\mathcal{F}\{3xe^{-2x}H(x-3)\}$ $\qquad \mathcal{F}\{f(x)\} = \int_{-\infty}^{+\infty} f(x)e^{-ikx}dx \qquad \mathcal{F}^{-1}\{F\} = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(k)e^{ikx}dk$

$\downarrow$

$\mathcal{F}\{3xe^{-2x}H(x-3)\} = 3\int_3^{\infty} xe^{-(2+ik)x}dx \qquad \int xe^{-\alpha x}dx = -\frac{x}{\alpha}e^{-\alpha x} - \frac{1}{\alpha^2}e^{-\alpha x} + c \qquad \int_3^{\infty} xe^{-\alpha x}dx = \left[-\frac{x}{\alpha}e^{-\alpha x} - \frac{1}{\alpha^2}e^{-\alpha x}\right]_3^{\infty} = e^{-3\alpha}\left(\frac{3}{\alpha}+\frac{1}{\alpha^2}\right)$

$\mathcal{F}\{3xe^{-2x}H(x-3)\} = 3e^{-3(2+ik)}\left(\frac{3}{2+ik} + \frac{1}{(2+ik)^2}\right) = 3e^{-3(2+ik)}\frac{7+3ik}{(2+ik)^2}$

$\mathcal{F}^{-1}\left\{\frac{\operatorname{sen}(3k-3)}{(3k-3)e^{3ik}}\right\} \qquad \int_{-a}^{a} e^{-ikx}dx = \frac{2\operatorname{sen}(ak)}{k} \qquad \mathcal{F}^{-1}F(k-k_0) = e^{ik_0x}f(x)$



- Fase $e^{-iak}$ in frequenza ⇒ traslazione nel tempo: $\mathcal{F}^{-1}e^{-iak}F(k) = f(x -$ .

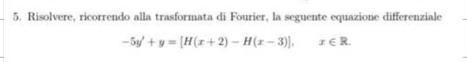
Scrivo

$$\frac{\sin(3k-3)}{(3k-3)e^{3ik}} = e^{-3ik}\cdot\frac{1}{3}\,\frac{\sin(3(k-1))}{(k-1)} = \frac{1}{6}\,e^{-3ik}\left[\frac{2\sin(3(k-1))}{(k-1)}\right].$$

$; e^{ix}, \mathbf{1}_{[-3,3]}(x)$ . Il fattore $e^{-3ik}$ trasla di $+3$ in $x$:

$$\boxed{\mathcal{F}^{-1}\left\{\frac{\sin(3k-3)}{(3k-3)e^{3ik}}\right\}(x) = \frac{1}{6}\,e^{i(x-3)}\,\mathbf{1}_{[0,6]}(x) = \frac{1}{6}\,e^{i(x-3)}\left[H(x)-H(x-6)\right]}$$

**ESERCIZIO n° 2**



5. Risolvere, ricorrendo alla trasformata di Fourier, la seguente equazione differenziale

$$-5y' + y = [H(x+2) - H(x-3)], \qquad x \in \mathbb{R}.$$

$-5y' + y = \left[H(x+2) - H(x-3)\right]$

1) DATI

$-5y'(x) + y(x) = f(x) \qquad f(x) = \left[H(x+2) - H(x-3)\right] \qquad x \in \mathbb{R}$ OSSERVA CHE $f$ È UN RETTANGOLO DI AMPIEZZA 5 CENTRO $x = \frac{1}{2}$

SCRIVO $f$ COME TRASLAZIONE DEL RETTANGOLO SIMMETRICO $\longrightarrow g(x) = H\left(x+\frac{5}{2}\right) - H\left(x-\frac{5}{2}\right) \qquad f(x) = g\left(x-\frac{1}{2}\right)$

2) TRASFORMATA DI $f$ CON LE FORMULE DELLA TABELLA

- $\rightarrow$ ONDA QUADRA PER $g(x) = H(x+a) - H(x-a)$ SI HA $g(x) = \frac{2\operatorname{sen}ak}{k}$ CON $a = \frac{5}{2}$ DUNQUE $\frac{\cancel{2}\operatorname{sen}\frac{5}{\cancel{2}}k}{k} \longrightarrow \frac{\operatorname{sen}5k}{k}$
- $\rightarrow$ TRASLAZIONE $f(x) = g(x-x_0) \longrightarrow f(k) = e^{-ikx_0}g(x)$ CON $x_0 = \frac{1}{2}$ QUINDI $f(k) = e^{-ik\frac{1}{2}}\frac{2\operatorname{sen}\frac{5}{2}k}{k}$

3) PASSO IN FOURIER

$\left.\begin{array}{l}\text{DERIVATA } \mathcal{F}\{y'\} = iky \\ \text{LINEARITÁ } \mathcal{F}\{-5y'+y\} = (-5ik+1)y\end{array}\right\} (1+5ik)y(k) = f(k) \longrightarrow y(k) = \frac{e^{-\frac{ik}{2}}\,2\operatorname{sen}\frac{5}{2}k}{k(1+5ik)}$

4) INVERSA CON LA CONVOLUZIONE DELLA TABELLA

- $\rightarrow$ COPPIA NOTA $\mathcal{F}\{e^{-ax}H(x)\} = \frac{1}{a+ik}$ CON $a = \frac{1}{5}$ ED USANDO SCALATURA/SOSTITUZIONE $x \mapsto x$ SI OTTIENE $\mathcal{F}^{-1}\left\{\frac{1}{1+5ik}\right\} = \frac{1}{5}e^{\frac{x}{5}}H(-x) = g_1(x)$ (È IL NUMERO DI GREEN DELL'OPERATORE $-5\frac{d}{dx}+1$)
- $\rightarrow$ CONVOLUZIONE

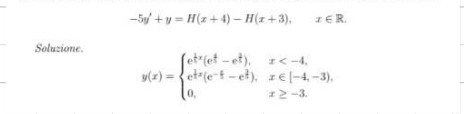
  $\mathcal{F}^{-1}\{F\cdot G\} = f\cdot g \qquad y = f\cdot\frac{1}{1+5ik} \Longrightarrow y = f\cdot g_1 \qquad y(x) = \int_{-\infty}^{+\infty} f(\xi)g_1(x-\xi)d\xi = \frac{1}{5}\int_{-\infty}^{+\infty} e^{\frac{(x-\xi)}{5}}H(\xi-x)d\xi$ IL GRADINO $H(\xi-x)$ IMPONE $\xi \geq x$ DISTINGUIAMO I CASI

5) INTEGRAZIONE A TRATTI

- $\rightarrow$ SE $x \geq 3$, L'INTERVALLO $[-2,3]$ NON CONTIENE PUNTI $\geq x \longrightarrow y(x) = 0$
- $\rightarrow$ SE $-2 \leq x < 3$ SI INTEGRA DA $\xi = x$ A $\xi = 3 \longrightarrow y(x) = \frac{1}{5}e^{\frac{x}{5}}\int_x^3 e^{-\frac{\xi}{5}}d\xi = e^{\frac{x}{5}}\left(e^{-\frac{x}{5}} - e^{\frac{3}{5}}\right) = 1 - e^{\frac{(x-3)}{5}}$
- $\rightarrow$ SE $x < -2$ SI INTEGRA DA $\xi = -2$ A $\xi = 3 \longrightarrow y(x) = \frac{1}{5}e^{\frac{x}{5}}\int_{-2}^3 e^{-\frac{\xi}{5}}d\xi = e^{\frac{x}{5}}\left(e^{-\frac{2}{5}} - e^{\frac{3}{5}}\right) = e^{\frac{(x+2)}{5}} - e^{\frac{(x+3)}{5}}$

$y(x) = \left[1 - e^{\frac{(x+3)}{5}}\right]H(3-x) + \left[e^{\frac{(x+3)}{5}} - 1\right]H(-2-x)$

## ESERCIZIO n° 3



5. Risolvere, ricorrendo alla trasformata di Fourier, la seguente equazione differenziale

$$-5y' + y = H(x+4) - H(x+3), \qquad x \in \mathbb{R}.$$

*Soluzione.*

$$y(x) = \begin{cases} e^{\frac{1}{5}x}(e^{\frac{4}{5}} - e^{\frac{3}{5}}), & x < -4, \\ e^{\frac{1}{5}x}(e^{-\frac{x}{5}} - e^{\frac{3}{5}}), & x \in [-4,-3), \\ 0, & x \geq -3. \end{cases}$$

$-5y'(x) + y(x) = U(x)$ $\qquad$ $U(x) = H(x+4) - H(x+3) = 1_{[-4,-3]}(x)$

1) IN FREQUENZA $\quad \mathcal{F}\{y'\} = i\omega y \longrightarrow (1-5i\omega)y(\omega) = U(\omega) \longrightarrow y = G(\omega)U \qquad G(\omega) = \frac{1}{1-5i\omega}$

2) GREEN DEL SISTEMA $\quad g = \mathcal{F}^{-1}\{G\}$ SODDISFA $-5g' + g = \delta$ SOLUZIONE A TRATTI E SALTO AL 0 (INTEGRO SU $(-\varepsilon,\varepsilon)$) $\quad -5(g(0^+) - g(0^-)) = 1 \longrightarrow g(0^-) - g(0^+) = \frac{1}{5}$

CON $-5g' + g = 0$ FUORI DA 0 OTTENGO $\quad g(x) = \frac{1}{5} e^{\frac{x}{5}} H(-x)$

3) CONVOLUZIONE $y = g \cdot U \qquad y(x) = \int_{\mathbb{R}} g(x-\xi) U(\xi) d\xi = \int_{-4}^{-3} \frac{1}{5} e^{\frac{(x-\xi)}{5}} H(\xi - x) d\xi$ $\quad$ L'HEAVISIDE $H(\xi - x)$ IMPONE $\xi \geq x$ QUINDI L'INTEGRALE È SU $[\max\{-4,x\}, -3]$

- → CASO $x < -4$

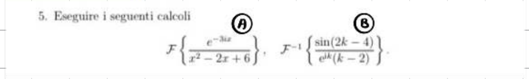
  $$y(x) = \frac{1}{5} e^{\frac{x}{5}} \int_{-4}^{-3} e^{-\frac{\xi}{5}} d\xi = e^{\frac{x}{5}} (e^{\frac{4}{5}} - e^{\frac{3}{5}})$$

- → CASO $-4 \leq x < -3$

  $$y(x) = \frac{1}{5} e^{\frac{x}{5}} \int_{-4}^{-3} e^{-\frac{\xi}{5}} d\xi = e^{\frac{x}{5}} (e^{-\frac{x}{5}} - e^{\frac{3}{5}})$$

- → CASO $x \geq -3$ INTERVALLO VUOTO $\longrightarrow y(x) = 0$

## ESERCIZIO n° 4



5. Eseguire i seguenti calcoli

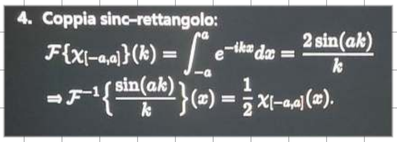
Ⓐ $\mathcal{F}\left\{\frac{e^{-3ix}}{x^2 - 2x + 6}\right\}$, $\quad$ Ⓑ $\mathcal{F}^{-1}\left\{\frac{\sin(2k-4)}{e^{ik}(k-2)}\right\}$.

LE PROPRIETÀ CHE USEREMO SARÀ:

TRASLAZIONE $\longrightarrow \mathcal{F}\{f(x-x_0)\}(k) = e^{-ikx_0} F(k)$

MODULAZIONE $\longrightarrow \mathcal{F}\{e^{ik_0x} f(x)\}(k) = F(k - k_0)$

DUALITÀ $\longrightarrow$ SE $\mathcal{F}\{g\} = f$ ALLORA $\mathcal{F}\{f\} = 2\pi g(-k)$



**4. Coppia sinc–rettangolo:**

$$\mathcal{F}\{\chi_{[-a,a]}\}(k) = \int_{-a}^{a} e^{-ikx} dx = \frac{2\sin(ak)}{k}$$
$$\Rightarrow \mathcal{F}^{-1}\left\{\frac{\sin(ak)}{k}\right\}(x) = \frac{1}{2}\chi_{[-a,a]}(x).$$

$\mathcal{F}\{f\}(k) = \int_{-\infty}^{+\infty} f(x) e^{-ikx} dx \qquad \mathcal{F}^{-1}\{F\}(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(k) e^{ikx} dk$

Ⓐ $\mathcal{F}\left\{\frac{e^{-3ix}}{x^2 - 2x + 6}\right\}(k)$

- → PASSO 1 (COMPLETA IL QUADRATO) $\quad x^2 - 2x + 6 = (x-1)^2 + 5$ $\quad$ DEFINISCO $g(x) = \frac{1}{x^2+5}$ $\quad f_0(x) = g(x-1) = \frac{1}{(x-1)^2+5}$ $\quad$ ALLORA $\quad \frac{e^{-3ix}}{x^2 - 2x + 6} = e^{-3ix} f_0(x)$
- → PASSO 2 (TRASFORMATA DI $g(x) = \frac{1}{x^2 + a^2}$ (CON $a > 0$)

  PER DUALITÀ SE $\mathcal{F}\{e^{-a|x|}\} = \frac{2a}{a^2+k^2}$ ALLORA $\mathcal{F}\left\{\frac{2a}{a^2+x^2}\right\}(k) = 2\pi e^{-a|x|}$ $\quad$ PER LINEARITÀ $\mathcal{F}\left\{\frac{1}{x^2+a^2}\right\}(k) = \frac{\pi}{a} e^{-a|k|}$ $\quad$ NEL NOSTRO CASO $a = \sqrt{5}$ $\quad \mathcal{F}\{g\}(k) = \frac{\pi}{\sqrt{5}} e^{-\sqrt{5}|k|}$
- → PASSO 3 (TRASLAZIONE $x \mapsto x-1$)

  $\mathcal{F}\{f_0\}(k) = e^{-ik} \longrightarrow \mathcal{F}\{g\}(k) = \frac{\pi}{\sqrt{5}} e^{-\sqrt{5}|k|}$
- → PASSO 4 (MODULAZIONE $e^{-i3x}$)

  $\mathcal{F}\{e^{-i3x} f_0(x)\}(k) \longrightarrow \mathcal{F}\{f_0\}(k+3)$

$$\left.\right\} \quad \mathcal{F}\left\{\frac{e^{-i3x}}{x^2 - 2x + 6}\right\}(k) = \frac{\pi}{\sqrt{5}} e^{-i(k+3)} \cdot e^{-\sqrt{5}|k+3|} = \frac{\pi\sqrt{5}}{5} e^{-i(k+3)} \cdot e^{-\sqrt{5}|k+3|}$$

ⓑ $\mathcal{F}^{-1}\left\{\frac{\text{sen}(2k-4)}{e^{ik}(k-2)}\right\}$

- → PASSO 1 (RISCRITTURA PULITA)

  $$\mathcal{F}(k) = \frac{\text{sen}(2k-4)}{e^{ik}(k-2)} = e^{-ik} \frac{\text{sen}(2(k-2))}{k-2} = e^{-ik} G(k-2)$$
- → PASSO 2 (ANTITRASFORMATA DI $G$)

  DALLA COPPIA SENC - RETTANGOLO CON $a = 2$ $\quad \mathcal{F}^{-1}\left\{\frac{\text{sen}(2k)}{k}\right\}(x) = \frac{1}{2} \chi_{[-2,2]}(x)$
- → PASSO 3 (TRASLAZIONE IN FREQUENZA)

  PER L'INVERSA, LA PROPRIETÀ DUALE È:

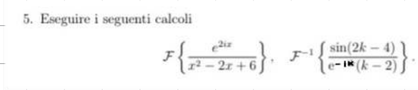
  $\mathcal{F}^{-1}\{H(k-k_0)\}(x) = e^{ik_0x} h(x) \qquad \mathcal{F}^{-1}\{G(k-2)\}(x) = e^{i2x} \frac{1}{2} \chi_{[-2,2]}(x)$
- → PASSO 4 (FATTORE $e^{-ik}$ SHIFT NEL TEMPO)

  DALLA PROPRIETÀ INVERSA $\quad \mathcal{F}^{-1}\{e^{-ik} Q(k)\}(x) = q(x-1) \qquad \mathcal{F}^{-1}\left\{\frac{\text{sen}(2k-4)}{e^{ik}(k-2)}\right\}(x) = \frac{1}{2} e^{2i(x-1)} \chi_{[-2,2]}(x-1)$
- → PASSO 5 (FORMA CON HEAVISIDE)

  $$\chi_{[-2,2]}(x-1) = \begin{cases} 1 & -2 \leq x-1 \leq 2 \iff -1 \leq x \leq x \\ 0 & \text{ALTRIMENTI} \end{cases} = H(x+1) - H(x-3)$$

$$f(x) = \frac{1}{2} e^{2i(x-1)} [H(x+1) - H(x-3)]$$

## ESERCIZIO n°5


5. Eseguire i seguenti calcoli

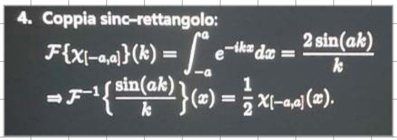
$$\mathcal{F}\left\{\frac{e^{2ix}}{x^2-2x+6}\right\}, \quad \mathcal{F}^{-1}\left\{\frac{\sin(2k-4)}{e^{-ik}(k-2)}\right\}.$$

LE PROPRIETÀ CHE USEREMO SARÀ:

TRASLAZIONE $\longrightarrow \mathcal{F}\{f(x-x_0)\}(k)=e^{-ikx_0}F(k)$

MODULAZIONE $\longrightarrow \mathcal{F}\{e^{ik_0x}f(x)\}(k)=F(k-k_0)$

DUALITÀ $\longrightarrow$ SE $\mathcal{F}\{g\}=f$ ALLORA $\mathcal{F}\{f\}=2\pi g(-k)$


**4. Coppia sinc-rettangolo:**

$$\mathcal{F}\{\chi_{[-a,a]}\}(k)=\int_{-a}^{a}e^{-ikx}dx=\frac{2\sin(ak)}{k}$$
$$\Rightarrow \mathcal{F}^{-1}\left\{\frac{\sin(ak)}{k}\right\}(x)=\frac{1}{2}\chi_{[-a,a]}(x).$$

$$\mathcal{F}\left\{\frac{e^{2ix}}{x^2-2x+6}\right\}$$

$\longrightarrow$ PASSO 1 (COMPLETA IL QUADRATO) $x^2-2x+6=(x-1)^2+5$ DEFINISCO $g(x)=\frac{1}{x^2+5}$ $f_0(x)=g(x-1)=\frac{1}{(x-1)^2+5}$ ALLORA $\frac{e^{2ix}}{x^2-2x+6}=e^{2ix}f_0(x)$

$\longrightarrow$ PASSO 2 (TRASFORMATA DI $g(x)=\frac{1}{x^2+a^2}$ (CON $a>0$)

PER DUALITÀ SE $\mathcal{F}\{e^{-a|x|}\}=\frac{2a}{a^2+k^2}$ ALLORA $\mathcal{F}\left\{\frac{2a}{a^2+x^2}\right\}(k)=2\pi e^{-a|k|}$ PER LINEARITÀ $\mathcal{F}\left\{\frac{1}{x^2+a^2}\right\}(k)=\frac{\pi}{a}e^{-a|k|}$ NEL NOSTRO CASO $a=\sqrt{5}$ $\mathcal{F}\{g\}(k)=\frac{\pi}{\sqrt{5}}e^{-\sqrt{5}|k|}$

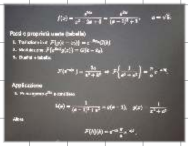
$\longrightarrow$ PASSO 3 (TRASLAZIONE $x\longmapsto x-1$)

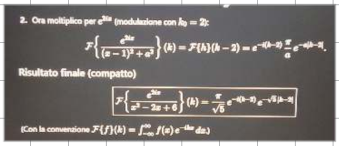
$\mathcal{F}\{f_0\}(k)=e^{-ik}\longrightarrow \mathcal{F}\{g\}(k)=\frac{\pi}{\sqrt{5}}e^{-\sqrt{5}|k|}$

$\longrightarrow$ PASSO 4 (MODULAZIONE $e^{2ik}$)

$\mathcal{F}\{e^{2ik}f_0(x)\}(k)\longrightarrow \mathcal{F}\{f_0\}(k-2)$

$$\mathcal{F}\left\{\frac{e^{2ix}}{x^2-2x+6}\right\}(k)=\frac{\pi}{\sqrt{5}}e^{-i(k-2)}\cdot e^{-\sqrt{5}|k-2|}=\frac{\pi\sqrt{5}}{5}e^{-i(k-2)}\cdot e^{-\sqrt{5}|k-2|}$$




**2. Ora moltiplico per $e^{2ix}$ (modulazione con $k_0=2$):**

$$\mathcal{F}\left\{\frac{e^{2ix}}{(x-1)^2+a^2}\right\}(k)=\mathcal{F}\{h\}(k-2)=e^{-i(k-2)}\frac{\pi}{a}e^{-a|k-2|}.$$

**Risultato finale (compatto)**

$$\boxed{\mathcal{F}\left\{\frac{e^{2ix}}{x^2-2x+6}\right\}(k)=\frac{\pi}{\sqrt{5}}e^{-i(k-2)}e^{-\sqrt{5}|k-2|}}$$

(Con la convenzione $\mathcal{F}\{f\}(k)=\int_{-\infty}^{\infty}f(x)e^{-ikx}dx$.)

$$\mathcal{F}^{-1}\left\{\frac{\text{sen}(2k-4)}{e^{-ik}(k-2)}\right\}$$

$\longrightarrow$ PASSO 1 (RISCRITTURA PULITA)

$\mathcal{F}(k)=\frac{\text{sen}(2k-4)}{e^{ik}(k-2)}=e^{-ik}\frac{\text{sen}(2(k-2))}{k-2}=e^{-ik}G(k-2)$

$\longrightarrow$ PASSO 2 (ANTITRASFORMATA DI G)

DALLA COPPIA SENC-RETTANGOLO CON $a=2$ $\mathcal{F}^{-1}\left\{\frac{\text{sen}(2k)}{k}\right\}(x)=\frac{1}{2}\chi_{[-2,2]}(x)$

$\longrightarrow$ PASSO 3 (TRASLAZIONE IN FREQUENZA)

PER L'INVERSA, LA PROPRIETÀ DUALE È:

$\mathcal{F}^{-1}\{H(k-k_0)\}(x)=e^{ik_0x}h(x)$ $\mathcal{F}^{-1}\{G(k-2)\}(x)=e^{i2x}\frac{1}{2}\chi_{[-2,2]}(x)$

$\longrightarrow$ PASSO 4 (FATTORE $e^{-ik}$ SHIFT NEL TEMPO)

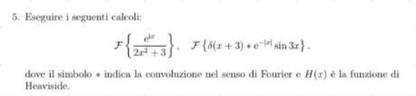
DALLA PROPRIETÀ INVERSA $\mathcal{F}^{-1}\{e^{-ik}G(k)\}(x)=g(x-1)$ $\mathcal{F}^{-1}\left\{\frac{\text{sen}(2k-4)}{e^{ik}(k-2)}\right\}(x)=\frac{1}{2}e^{2i(x-1)}\chi_{[-2,2]}(x-1)$

$\longrightarrow$ PASSO 5 (FORMA CON HEAVISIDE)

$$\chi_{[-2,2]}(x-1)=\begin{cases}1 & -2\le x-1\le 2 \iff -1\le x\le x\\ 0 & \text{ALTRIMENTI}\end{cases}=H(x+1)-H(x-3)$$

$$f(x)=\frac{1}{2}e^{2i(x-1)}\left[H(x+1)-H(x-3)\right]$$

## ESERCIZIO n°6


5. Eseguire i seguenti calcoli:

$$\mathcal{F}\left\{\frac{e^{ix}}{2x^2+3}\right\}, \quad \mathcal{F}\left\{\delta(x+3)*e^{-|x|}\sin 3x\right\}.$$

dove il simbolo $*$ indica la convoluzione nel senso di Fourier e $H(x)$ è la funzione di Heaviside.

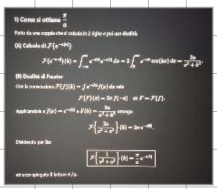
$$\mathcal{F}\left\{\frac{e^{ix}}{2x^2+3}\right\}$$

$\longrightarrow$ COPPIA NOTA $\mathcal{F}\left\{\frac{1}{x^2+a^2}\right\}(k)=\frac{\pi}{a}e^{-a|k|}$ ($a>0$)

$\longrightarrow$ QUINDI $\frac{1}{2x^2+3}=\frac{1}{2}\cdot\frac{1}{x^2+\frac{3}{2}}\longrightarrow a=\sqrt{\frac{3}{2}}$ $\mathcal{F}\left\{\frac{1}{2x^2+3}\right\}(k)=\frac{1}{2}\cdot\frac{\pi}{a}e^{-a|k|}=\frac{\pi}{\sqrt{6}}e^{-\sqrt{\frac{3}{2}}|k|}$

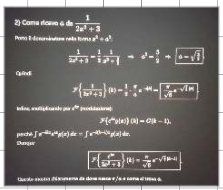
$\longrightarrow$ MOLTIPLICAZIONE $e^{ix}\longrightarrow$ SPOSTO IN FREQUENZA $\mathcal{F}\{e^{ik_0x}f(x)\}(k)=F(k-k_0)$ CON $k_0=1$

$\mathcal{F}\left\{\frac{e^{ix}}{2x^2+3}\right\}(k)=\frac{\pi}{\sqrt{6}}e^{-\sqrt{\frac{3}{2}}|k-1|}$


1) Come si ottiene $\frac{\pi}{a}$

(A) Calcolo di $\mathcal{F}\{e^{-a|x|}\}$

(B) Dualità di Fourier

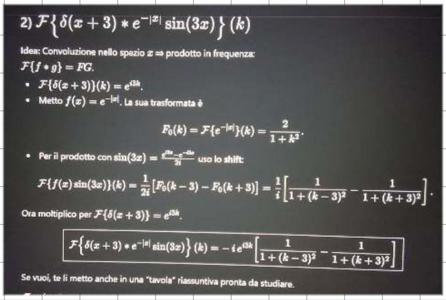

2) Come ricavo $a$ da $\frac{1}{2x^2+3}$

Quindi

Dunque

$$\mathcal{F}\{\delta(x+3)\cdot e^{-|x|}\,\text{sen}(3x)\}(k)$$

CONVOLUZIONE NELLO SPAZIO $\longrightarrow$ PRODOTTO IN FREQUENZA


2) $\mathcal{F}\left\{\delta(x+3)*e^{-|x|}\sin(3x)\right\}(k)$

Idea: Convoluzione nello spazio $x\Rightarrow$ prodotto in frequenza:
$\mathcal{F}\{f*g\}=FG.$

- $\mathcal{F}\{\delta(x+3)\}(k)=e^{i3k}.$
- Metto $f(x)=e^{-|x|}$. La sua trasformata è

$$F_0(k)=\mathcal{F}\{e^{-|x|}\}(k)=\frac{2}{1+k^2}.$$

- Per il prodotto con $\sin(3x)=\frac{e^{i3x}-e^{-i3x}}{2i}$ uso lo shift:

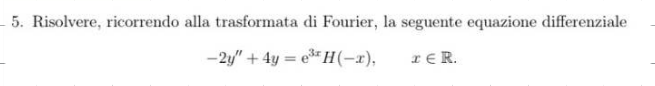
$$\mathcal{F}\{f(x)\sin(3x)\}(k)=\frac{1}{2i}[F_0(k-3)-F_0(k+3)]=\frac{1}{i}\left[\frac{1}{1+(k-3)^2}-\frac{1}{1+(k+3)^2}\right].$$

Ora moltiplico per $\mathcal{F}\{\delta(x+3)\}=e^{i3k}.$

$$\boxed{\mathcal{F}\left\{\delta(x+3)*e^{-|x|}\sin(3x)\right\}(k)=-ie^{i3k}\left[\frac{1}{1+(k-3)^2}-\frac{1}{1+(k+3)^2}\right]}$$

Se vuoi, te li metto anche in una "tavola" riassuntiva pronta da studiare.

## ESERCIZIO n°7



5. Risolvere, ricorrendo alla trasformata di Fourier, la seguente equazione differenziale

$$-2y'' + 4y = e^{3x}H(-x), \qquad x \in \mathbb{R}.$$

$-2y'' + 4y = e^{3x} H(-x) \quad x \in \mathbb{R}$

1) TRASFORMATA DI FOURIER

CON $\mathcal{F}\{y''\} = -(k^2)y$ $\quad (2k^2+4)\,y(k) = f(k)$ $\quad f(x) = e^{3x}H(-x)$ DALLE FORMULE DELL'IMPULSO ESPONENZIALE $f(k) = \int_{-\infty}^{0} e^{(3-ik)}dx = \frac{1}{3-ik}$ QUINDI $y(k) = \frac{1}{(2k^2+4)(3-ik)} = \frac{1}{2}\cdot\frac{1}{(k^2+2)(3-ik)}$

2) INVERSA (CONVOLUZIONE)

PRODOTTO IN K $\longrightarrow$ CONVOLUZIONE IN X DALLE COPPIE NOTE $\mathcal{F}^{-1}\left\{\frac{1}{3-ik}\right\} = e^{3x}H(-x) =: g(x)$ $\quad \mathcal{F}^{-1}\left\{\frac{1}{k^2+2}\right\} = \frac{1}{2\sqrt{2}}e^{-\sqrt{2}|x|}$ ALLORA $\mathcal{F}^{-1}\left\{\frac{1}{2(k^2+2)}\right\} = \frac{1}{4\sqrt{2}}e^{-\sqrt{2}|x|} =: h(x)$ QUINDI $y = g \cdot h = \frac{1}{4\sqrt{2}}\int_{-\infty}^{0} e^{3\xi}\cdot e^{-\sqrt{2}|x-\xi|}d\xi$

3) VALUTAZIONE DELL'INTEGRALE

$\longrightarrow$ CASO $x \geq 0$

$|x-\xi| = x-\xi$ PER OGNI $\xi \leq 0$ $\quad y(x) = \frac{e^{-\sqrt{2}x}}{4\sqrt{2}}\int_{-\infty}^{0} e^{(3+\sqrt{2})\xi}d\xi = \frac{e^{-\sqrt{2}x}}{4\sqrt{2}(3+\sqrt{2})}$ RAZIONALIZZO I COEFFICIENTI $\frac{e^{-\sqrt{2}x}}{4\sqrt{2}(3+\sqrt{2})}\cdot\frac{(3+\sqrt{2})}{(3+\sqrt{2})} = \frac{e^{-\sqrt{2}x}\cdot\sqrt{2}}{8(3+\sqrt{2})} = \frac{e^{-\sqrt{2}x}\cdot\sqrt{2}(3-\sqrt{2})}{56} = \boxed{\frac{e^{-\sqrt{2}x}\cdot 3\sqrt{2}-2}{56} \quad x>0}$

$\longrightarrow$ CASO $x<0$

DIVIDO L'INTEGRALE $\int_{-\infty}^{0} = \int_{-\infty}^{x} + \int_{x}^{0}$

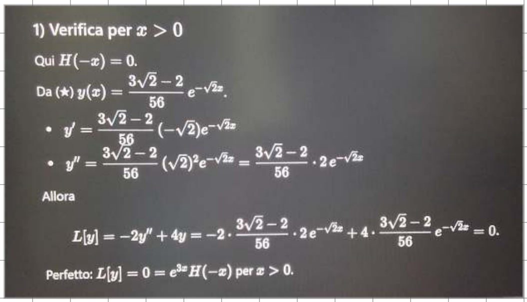
PER $\xi \leq x : |x-\xi| = x-\xi \longrightarrow \int_{-\infty}^{x} e^{3\xi}\cdot e^{-\sqrt{2}(x-\xi)}d\xi = \frac{e^{3x}}{3+\sqrt{2}}$

PER $x \leq \xi \leq 0 : |x-\xi| = \xi - x \longrightarrow \int_{x}^{0} e^{3\xi}\cdot e^{-\sqrt{2}(\xi-x)}d\xi = \frac{e^{\sqrt{2}x}-e^{3x}}{3-\sqrt{2}}$ SOMMO E METTO IL FATTORE $\frac{1}{4\sqrt{2}}$

$y(x) = \frac{1}{4\sqrt{2}}\left[\frac{e^{3x}}{3+\sqrt{2}} + \frac{e^{\sqrt{2}x}-e^{3x}}{3-\sqrt{2}}\right] = -\frac{1}{14}e^{3x} + \frac{e^{\sqrt{2}x}}{4\sqrt{2}(3-\sqrt{2})}$ RAZIONALIZZO I COEFFICIENTI $\frac{1}{4\sqrt{2}(3+\sqrt{2})}\cdot\frac{(3+\sqrt{2})}{(3+\sqrt{2})} = \frac{1\cdot\sqrt{2}}{8(3+\sqrt{2})} = \frac{1\cdot\sqrt{2}(3-\sqrt{2})}{56} = \frac{1\cdot 3\sqrt{2}-2}{56}$ QUINDI $\boxed{y(x) = \frac{3\sqrt{2}+2\cdot e^{\sqrt{2}x}}{56} - \frac{1}{14}e^{3x} \quad x \leq 0}$

SOLUZIONE

$$\boxed{y(x) = \begin{cases} \frac{e^{-\sqrt{2}x}\cdot 3\sqrt{2}-2}{56} & x>0 \\ \frac{3\sqrt{2}+2\cdot e^{\sqrt{2}x}}{56} - \frac{1}{14}e^{3x} & x \leq 0 \end{cases}}$$



### 1) Verifica per $x > 0$

Qui $H(-x) = 0$.

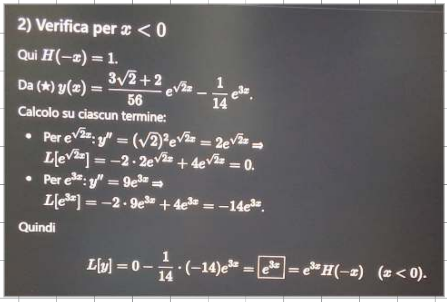
Da (★) $y(x) = \frac{3\sqrt{2}-2}{56}e^{-\sqrt{2}x}$.

- $y' = \frac{3\sqrt{2}-2}{56}(-\sqrt{2})e^{-\sqrt{2}x}$
- $y'' = \frac{3\sqrt{2}-2}{56}(\sqrt{2})^2e^{-\sqrt{2}x} = \frac{3\sqrt{2}-2}{56}\cdot 2e^{-\sqrt{2}x}$

Allora

$$L[y] = -2y'' + 4y = -2\cdot\frac{3\sqrt{2}-2}{56}\cdot 2e^{-\sqrt{2}x} + 4\cdot\frac{3\sqrt{2}-2}{56}e^{-\sqrt{2}x} = 0.$$

Perfetto: $L[y] = 0 = e^{3x}H(-x)$ per $x > 0$.



### 2) Verifica per $x < 0$

Qui $H(-x) = 1$.

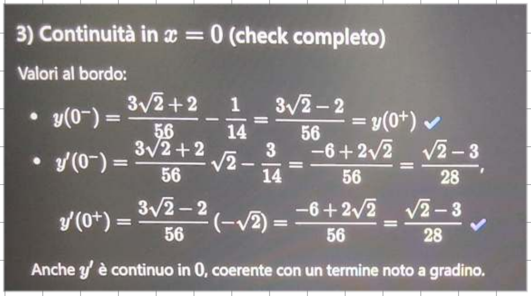
Da (★) $y(x) = \frac{3\sqrt{2}+2}{56}e^{\sqrt{2}x} - \frac{1}{14}e^{3x}$.

Calcolo su ciascun termine:

- Per $e^{\sqrt{2}x}$: $y'' = (\sqrt{2})^2e^{\sqrt{2}x} = 2e^{\sqrt{2}x} \Rightarrow$ $L[e^{\sqrt{2}x}] = -2\cdot 2e^{\sqrt{2}x} + 4e^{\sqrt{2}x} = 0$.
- Per $e^{3x}$: $y'' = 9e^{3x} \Rightarrow$ $L[e^{3x}] = -2\cdot 9e^{3x} + 4e^{3x} = -14e^{3x}$.

Quindi

$$L[y] = 0 - \frac{1}{14}\cdot(-14)e^{3x} = \boxed{e^{3x}} = e^{3x}H(-x) \quad (x<0).$$



### 3) Continuità in $x = 0$ (check completo)

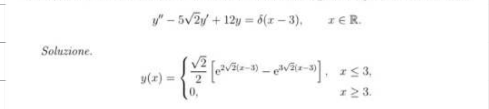
Valori al bordo:

- $y(0^-) = \frac{3\sqrt{2}+2}{56} - \frac{1}{14} = \frac{3\sqrt{2}-2}{56} = y(0^+)$ ✓
- $y'(0^-) = \frac{3\sqrt{2}+2}{56}\sqrt{2} - \frac{3}{14} = \frac{-6+2\sqrt{2}}{56} = \frac{\sqrt{2}-3}{28}$,

  $y'(0^+) = \frac{3\sqrt{2}-2}{56}(-\sqrt{2}) = \frac{-6+2\sqrt{2}}{56} = \frac{\sqrt{2}-3}{28}$ ✓

Anche $y'$ è continuo in 0, coerente con un termine noto a gradino.

## ESERCIZIO n°8



5. Risolvere, ricorrendo alla trasformata di Fourier, la seguente equazione differenziale

$$y'' - 5\sqrt{2}y' + 12y = \delta(x-3), \qquad x \in \mathbb{R}.$$

*Soluzione.*

$$y(x) = \begin{cases} \frac{\sqrt{2}}{2}\left[e^{2\sqrt{2}(x-3)} - e^{3\sqrt{2}(x-3)}\right], & x \leq 3, \\ 0, & x \geq 3. \end{cases}$$

$y'' - 5\sqrt{2}y' + 12y = \delta(x-3) \quad x \in \mathbb{R}$

$\longrightarrow$ TRASFORMO

USO $\mathcal{F}\{y'\} = ikY$ $\quad \mathcal{F}\{y''\} = (ik)^2Y$ E $\mathcal{F}\{\delta(x-x_0)\} = e^{-ikx_0}$

POSTO $Y(k) = \mathcal{F}\{y\}$

$\left[(ik)^2Y - 5\sqrt{2}(ik) + 12\right]Y(k) = e^{-i3k} \xrightarrow{\text{QUINDI}} Y(k) = \frac{e^{-i3k}}{(ik-2\sqrt{2})(ik-3\sqrt{2})} = \frac{e^{-i3k}}{\sqrt{2}}\left[\frac{1}{ik-2\sqrt{2}} - \frac{1}{ik-3\sqrt{2}}\right]$

$\longrightarrow$ ANTITRASFORMO

$\mathcal{F}^{-1}\left\{\frac{1}{a+ik}\right\} = e^{-ax}H(x)$ $\quad \mathcal{F}^{-1}\left\{\frac{1}{a-ik}\right\} = e^{ax}H(-x)$ $\quad a>0$ $\quad \mathcal{F}^{-1}\left\{\frac{1}{ik-a}\right\} = -e^{ax}H(-x)$

$y(x) = \frac{1}{\sqrt{2}}\left[-e^{2\sqrt{2}(x-3)}H(3-x) + e^{3\sqrt{2}(x-3)}H(3-x)\right] \xrightarrow{\text{FORMA COMPATTA}} y(x) = \begin{cases} \frac{\sqrt{2}}{2}\left(e^{2\sqrt{2}(x-3)} - e^{3\sqrt{2}(x-3)}\right) & x \leq 3 \\ 0 \quad x \geq 3 \end{cases}$